

Bash scripts
from Learn Linux TV

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July 12, 2026

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Chapter 1

Hello World

1.1 To create a bash script

First, we'll have to create a file with a .sh extension. Then we can edit it in with whatever command we want.

1.1: 1-bash.sh

```
1 ls
2 pwd
```

The way bash scripts are execute is top to bottom.

1.2 how to run it?

First we'll have to change the script properties from the terminal.

1.2: Terminal

```
1 sudo chmod +x 1-bash.sh
```

This makes the script executable. Then we just have to execute it from the terminal:

1.3: Terminal

```
1 ./1-bash.sh
```

1.3 Good practice for new scripts

What we are using here is a "shebang". This tells the interpreter whether they should be the ones executing this script. Otherwise, they should call the specified interpreter to execute the script.

1.4: 1-bash.sh

```
1 #!/bin/bash
2
3 ls
4 pwd
```

1.4 Comments

You can add comments to you script using '#' followed by text.

1.5 Hello World!

So to write anything out, you use the echo command. Here we'll do what we always do at the beginning and write our hello world.

1.5: 1-bash.sh

```
1 #!/bin/bash
2
3 echo "Hello World"
```

Chapter 2

Variables

2.1 Definition and usage

defining a variable is completely different than using it. To use it, you have to reference the variable similar to C.

2.1: 2-bash.sh

```
1 #!/bin/bash
2
3 # Do not use spaces between the variable, '=' and value
4 myName="John Doe"
5
6 echo $myName
```

2.2 Variables are temporary

This type of variable definition would not survive longer than the session. What I mean by session is that once you close the terminal or logout that specific user, the variables are going to be erased. This means that defining the variables using scripts can guarantee that needed variables are going to be available for the script itself.

2.3 String interpolation available

If you know about string interpolation, you know the dollar sign is the standard way to reference a variable inside a string and this mechanism also works in bash.

2.2: 2-bash.sh

```
1 #!/bin/bash
2
3 myName="John"
4 myAge="99"
5
6 echo "Hello, my name is $myName and I'm $myAge years old"
```

Also, it is required to use double quotes as that is the type of string that bash understands has the capabilities for string interpolation. If you use single strings, the variable won't be referenced and instead, "\$myVariable" would be directly displayed

Also, you can use \" if you want to display a double quote character in your string.

2.4 Store command outputs in variables

There is this concept known as a subshell. It essentially executes a command inside a command. This allows us to store the output of a command in a variable.

2.3: 2-bash.sh

```
1 #!/bin/bash
2
3 files=$(ls)
4
5 echo $files
```

2.5 Environment variables

There are already some predefined variables. If you wish to see them, you can use the env command

2.4: Terminal

```
1 env
```

Chapter 3

Math in Bash

3.1 how to run basic math operators

For you to make basic operations like $10 + 9$, you need to use the “evaluate expression” command.

3.1: 3-bash.sh

```
1 #!/bin/bash
2
3 #the space between value and operator is mandatory
4 expr 30 + 10
5
6 expr 30 - 10
7
8 expr 30 / 10
```

3.2 Wild cards

You might think that doing a multiplication is as simple as doing “`expr 10 * 10`” but you’d be wrong. This is because the asterisk character is a wild card. Wild cards are essentially special characters that return specific files depending on their own criteria. For example, `*.log` essentially looks for all files that end in `.log` and returns them to the command. this means that a command might perform an operation on all files that end in `.log` as oposed of only doing one operation. Using ‘`*`’ without anything else means to literally use everything.

So how do we solve this issue? we have to use the escape character

3.2: 3-bash.sh

```
1 #!/bin/bash
2
3 expr 30 \* 10
```

3.3 Math with variables

Going back to the previous chapter, we can use variables with values and perform operations

3.3: 3-bash.sh

```
1 #!/bin/bash
2
3 myVar=100
4
5 expr $myVar + 50
```


3.4 Arithmetic expansion

Now we can continue using `expr` or use the more modern and preferred way of doing math in general which is using arithmetic expansions. This don't have space constraints nor require flags.

3.4: 3-bash.sh

```
1 #!/bin/bash
2
3 # because the arithmetic expansion is a command itself, we must implement with a subshell
4 myVar=$((20 * 100))
5
6 echo $myVar
```

As we'll see later, using the subshell isn't required for if conditionals.

Chapter 4

If statements

4.1 If condition

If statements are weird in bash. First of all, bash is very space sensitive so manage with care. Then the syntax is very atypical.

4.1: 4-bash.sh

```
1 #!/bin/bash
2
3 myNum=200
4
5 # All spaces are mandatory!
6
7 if [ $myNum -eq 200 ]
8 then
9     echo "the condition is true"
10 fi # if backwards. bruh
```

Now, you might be wondering why are we using a command flag inside our if statement? And the reason for that is because actually, the square bracket is a command itself. It's an abbreviation of the test command. The -eq flag stands for equals to and expects an argument afterward. The closing square bracket is just expected by the alias definition of the test command.

4.2 A more modern approach for number comparison

Now, if you wanted to use your typical ==, !=, >, etc operators, you would use a double parentheses command. This is as modern as the year 1997 can be labeled modern, which is not at all. So use this if you want a more normal experience.

4.2: 4-bash.sh

```
1 #!/bin/bash
2 myVar=200
3 if (($myVar==200))
4 then
5     echo "myVar is equal to 200"
6 fi
```

4.3 Else statement

4.3: 4-bash.sh

```
1 #!/bin/bash
2 myVar=200
3 if (($myVar==200))
4 then
5     echo "myVar is equal to 200"
6 else
7     echo "myVar is not equal to 200"
8 fi
```

4.4 Not operator

4.4: 4-bash.sh

```
1 #!/bin/bash
2 myVar=200
3
4 # You can add an exclamation mark just like any other language, without the need of
   further parentheses
5 if ((!$myVar==200))
6 then
7     echo "myVar is not equal to 200"
8 else
9     echo "myVar is equal to 200"
10 fi
```

4.5 -f flag

Since the ‘[]’ operator is a bash command, it can interact with directories, files and such. This means we can compare against files or directories directly. The ‘-f’ flag is just one way to do this, where it checks whether a path exists AND is a file. If you just wanted to see if it existed as a file or directory, you’d use the -e flag.

4.5: 4-bash.sh

```
1 #!/bin/bash
2
3 # Does the 1-bash.sh file exist?
4 if [ -f ./1-bash.sh ]
5 then
6     echo "the 1-bash.sh file exists"
7 else
8     echo "the 1-bash.sh file doesn't exist"
9 fi
```

4.6 A more “practical” example

We want to check if a command exists, if it does run it, otherwise install it and then run it.

4.6: 4-bash.sh

```
1 #!/bin/bash
2
3 # it's good practice to use double quotes in case the path has spaces. if they do, bash
  might try to run a command afterwards leading to errors.
4 command="/usr/bin/htop"
5
6 # Yes, you also have to use double quotes in here for the same reason
7 if [ -f "$command" ]
8 then
9     echo "$command exists"
10 else
11     echo "$command does not exist, installing it"
12     # && means 'AND' execute this other command but only if the previous one executed
    without issues
13     sudo apt-get update && sudo apt-get upgrade -y && sudo apt install htop
14 fi
15
16 #execute command
17 $command
```

4.7 The better way of doing it

We used the test command to check whether a file existed in our system. However, this is a very general way of doing it, the shell has built in ways of managing this specific situation. We want to check not if a file exists but if a command exists. So we'll use a Shell builtin known as a command. Shell builtins are like any other commands but they are baked directly into the shell as opposed to being binary files. A common example of this would be the cd command. If you wish to know whether a command is a shell builtin, use the type command.

4.7: 4-bash.sh

```
1 #!/bin/bash
2
3 myCommand=htop
4
5 # command looks if the specified command exists. If it does, the -v flag outputs the
  direction of the command. or just the name of it if it's a shell builtin. Otherwise,
  it won't return anything
6 if command -v $myCommand
7 then
8     echo "$myCommand is installed, running it"
9 else
10    echo "$myCommand not installed, installing it"
11    sudo apt-get update && sudo apt-get upgrade -y && sudo apt install $myCommand -y
12 fi
13
14 $myCommand
```

The reason why you wouldn't want to use which is because the which command doesn't contemplate shell builtin commands so it might not work with every command. Also command tends to be faster since it doesn't require to open up a file and do many processes but is baked into the running process. known as the shell.

4.8 Logic operators

We already saw the AND operator. Naturally, it can also be used inside the if statement argument.

4.8: 4-bash.sh

```
1 #!/bin/bash
2
3 # Either one must exit with 0 for the if statement to run its inner code
4 if command -v top || command -v htop
5 then
6     echo "It is possible to profile the processes"
7 else
8     echo "no terminal tools were found to profile processes"
9 fi
```

4.9 Else if

Of course any well self respecting language would have the inclusion of the humble else if.

4.9: 4-bash.sh

```
1 #!/bin/bash
2
3 myCommand="cd"
4
5 if which $myCommand
6 then
7     echo "$myCommand exists"
8 elif command -v $myCommand
9 then
10    echo "$myCommand exists"
11 else
12    echo "$myCommand doesn't exist"
13 fi
```

4.10 Ok but how does the if statement actually work?

The if statements in bash scripting don't work with your typical boolean values, they are not expecting a true nor a false. What they work with is exit codes. If a command ran successfully, it will return a 0 exit code otherwise, any other number. So 0 behaves like true, any other value behaves like false. This is the opposite of how you'd expect it to work. More on exit codes next chapter.

Chapter 5

Exit codes

5.1 Check the exit code of the previous command

you can check the exit code of previous commands using the dollar sign. If it's 0, it means success. Otherwise, it means failure. The number might give you a clue on the failure actually was about.

5.1: 5-bash.sh

```
1 #!/bin/bash
2
3 package=htop
4
5 sudo apt install $package
6
7 if (($?==0))
8 then
9     echo "the installation of $package was successful"
10    echo "you can find it here:"
11    which $package
12 else
13     echo "failed to install $package"
14 fi
```

5.2 Redirects

Redirects allow us to save the output of a command directly into a file.

5.2: 5-bash.sh

```
1 #!/bin/bash
2
3 package=htop
4
5 # Using a single greater than symbol would result in the specified file being overwritten
6 # Additionally, the output won't be displayed on the cli. It will be fully redirected*
7 # *It won't in some cases but we'll see that later
8
9 sudo apt install $package > package_install.log
10
11 # Using 2 greater than symbols would add the output at the end of the file.
12 sudo apt install $package >> package_install.log
```

Something to consider is that since the file doesn't follow a strict path, the script will create the file in your current path where you executed the script.

5.3 Careful with \$?

Something you have to remember is that `$?` will only give you the exit code of the last command so if you want to check the first command, you have to store the value in another variable

5.3: 5-bash.sh

```
1 #!/bin/bash
2
3 myPath=/etc
4
5 if [ -d myPath ]
6 then
7     echo "The $myPath directory exist"
8 else
9     echo "The $myPath directory doesn't exist"
10 fi
11
12 # This will always print a 0 because it checks the exit code of the echoes which are
13 # always 0.
14 echo "The exit code is $?"
```

5.4 Manually exit

If you didn't want to store the exit value in a variable, which can get convoluted real quick, you can manually exit your script with a custom code.

5.4: 5-bash.sh

```
1 #!/bin/bash
2
3 echo "Hello world"
4 exit 199
5
6 # Anything after the exit command will be ignored.
```

5.5 The final version of the 5-bash.sh script

5.5: 5-bash.sh

```
1 #!/bin/bash
2
3 myPath=/etc
4
5 if [ -d $myPath ]
6 then
7     echo "The $myPath directory exists"
8     exit 0
9 else
10    echo "The $myPath directory doesn't exist"
11    exit 1
12 fi
```

Chapter 6

While loops

6.1 Basic while loop

6.1: 6-bash.sh

```
1 #!/bin/bash
2
3 myVar=1
4
5 # While the condition remains true, it will continue
6 while (($myVar<=10))
7 do
8     # ++ exists but apparently it doesn't work if the var starts at 0 which defeats its
8     # purpose more often than not.
9     ((myVar+=1))
10 done
11
12 # Because it's <=, it will make a final pass on the body making myVar = 11
13 echo $myVar
```

6.2 The sleep command

If we wanted to give our loop or realistically any commands in our scripts a small delay, we could use the sleep command.

6.2: 6-bash.sh

```
1 #!/bin/bash
2
3 while [ -f ./testfile ]
4 do
5     echo "File exists"
6     sleep 0.5
7 done
8
9 echo "The file was deleted at $(date)"
```

In a more realistic scenario, the delay would be higher or the implementation might be entirely different.

Chapter 7

How updates work in linux

7.1 Repositories

Lets start demystifying how linux installation works by starting with the place where our updates come from, repositories. **Repositories** are servers that contain software that is constantly being updated by the maintainers and where we can download our updates from.

There is also the concept of **mirrors** which are literal copies of the repositories as they have the same content (thus the name mirrors) but are located in different places. This is useful because you can choose mirrors that are closer to you to improve download speeds

If you want to see where the repository list is located in your computer, you can check `/etc/apt/source.list.d/`

7.2 Starting the workflow: apt update

Each distro has their own way of updating software but we'll be looking at debian based distros for this one. The first thing we do is to “**update**”:

7.1: Terminal

```
1 apt update
```

What “apt update” does is that it checks the list of repositories and downloads a type of metadata known as **indexes**. These indexes contain data that apt can later use to see if there is a pending update for a software, how to solve dependencies, where specifically do we install the files from (inside the repository path), etc ...

You see, The repositories are divided into main paths, one known as “dists/” which contains metadata. Using metadata is also useful because now we don't have to check entire gigabytes of files to see if there is an update available. Then we have the “pool/” path which contains the actual files to be downloaded.

You can check the indexes at `/var/lib/apt/lists`. Usually, the files are stored in compressed files.

7.3 Actually installing: apt upgrade

Now we move into the installation workflow. “[apt upgrade](#)” starts by checking those indexes and loading them into memory. Then it loads some sort of metadata of currently installed packages into memory as well which can be found at [/var/lib/dpkg/status](#). This means that we now can compare the metadata of the repositories vs the ones corresponding to locally installed packages to see which require updates.

Something to consider is that their installation order must follow a dependency order because we cannot install a package that depends on another one. Because of this, apt builds a dependency graph, specifically a **Directed Acyclic Graph (DAG)**. DAGs are essentially like any other graph that have connections with directions, and are acyclic which means that no direction can result in the creation of a loop.

Once we know which software must be updated and in which order, we can start downloading the packages. All packages are downloaded into a cache directory at [/var/cache/apt/archives/](#). This is so if there was to be an internet outage or a package is found corrupt, the interruption wouldn’t reset the entire progress back to 0. This implies that the cache is also checked before doing the installation.

Finally, we move into the actual installation. Apt, isn’t responsible for it since it cannot process .deb files so it calls dpkg to handle that. **dpkg** is the package installer for .deb files. It will install all files in the cache respecting the order set by the DAG. We won’t cover how dpkg works, for now...

7.4 The alternative: apt dist-upgrade

Sometimes, a package will need a new dependency that wasn’t contemplated. Apt by default won’t add any new packages on an update. Because of this, the packages will be labeled as “held back”. To force apt to install new packages and resolve all sorts of dependency issues it may encounter, you can use “[apt dist-upgrade](#)”.

Chapter 8

Update script

8.1 A script for all distros

Distro might have different update mechanisms. If you deal with multiple updates, this script can be a one fits all solution.

8.1: 8-bash.sh

```
1 #!/bin/bash
2
3 if [ -d /etc/pacman.d ]
4 then
5     # for archzzz
6     sudo pacman -Syu
7 elif [ -d /etc/apt ]
8 then
9     # for debian BASED!!!! literally
10    sudo apt update && sudo apt upgrade
11 fi
```

8.2 A better way of writing the script

the script relies on the specific software to be installed. However there can be nutheads that manage to install pacman in debian based distros. For this nutheads, we need to check the distro installed opposed to the update manager.

8.2: 8-bash.sh

```
1 #!/bin/bash
2
3 release_file=/etc/os-release
4 # we'll see what grep and other commands do after this chapter
5 if grep -q "Arch" $release_file
6 then
7     sudo pacman -Syu
8 elif grep -q "ubuntu" $release_file
9 then
10    sudo apt update && sudo apt upgrade
11 fi
```

Chapter 9

The find command

9.1 What is it?

This command allows us to search for files or directories in specified paths with different options to narrow down the specifics of what we are looking for.

9.1: 9-bash.sh

```
1 #!/bin/bash
2
3 # First we have to specify the path where we want to conduct the search
4 # Then we can select an option; to specify what the name of the file should look like in
   this case
5 # Search will be recursive so subdirectories will also be processed
6 find /home/jonathan -name *.txt
7
8 # from home path path
9 find /home/jonathan -name Documents
```

9.2 The find command is very extensive

There are many options that you can use in order to customize your search without taking into account we can also use wild cards in combination. First, lets take a look at the '-type' test expression.

9.2: 9-bash.sh

```
1 #!/bin/bash
2
3 # the type expression specifies what type of file we want to search for. In linux,
   everything is a file, even directories.
4 # there are many options for type but the 2 most important ones are 'f' for file and 'd'
   for directory
5 find /home/jonathan/ -name Documents -type d
```

9.3 Execute commands if file is found

There is the ‘exec’ action expression which allows you to execute any other command if the find command successfully finds a file (exit code = 0).

9.3: 9-bash.sh

```
1 #!/bin/bash
2
3 # 1. we look for a file named trash in current dir
4 # 2. The output(s) of the find command will be redirected as input to the exec command.
   Where? at the '{}'
5 # So -exec rm ./trash +
6 # 3. The rm command will have to start new instances and end them over and over.
7 # This can hit performance so we can use '+' to say: ‘Don't stop the process, continue
   with the next output given by find!’
8 find ./ -name trash -type f -exec rm {} +
```

Alternatively, you could use ‘\;’ to terminate (which is mandatory for -exec) if you know there is only one output from find.

9.4 A more practical example of the find command

say you have a list of images you wish to keep only visible to your user. You'd have to run:

9.4: Terminal

```
1 chmod 600 -R ./Pictures/
```

We don't want the file to be executable, that's why we go with 600. There's a problem with this approach whatsoever. The “Pictures” directory will also be marked as rw-. This is bad because in order for anyone to access a directory, it must have the executable option on. This means that we need to apply chmod 600 only to files and ignore directories and subdirectories. We could fix this using:

9.5: Terminal

```
1 chmod u+x -R ./Pictures/
```

But if we were dealing with files, making them executable introduces vulnerabilities or the chance of bricking the system. Most likely there is a way to achieve our purpose with the chmod command itself. However this is a perfect opportunity to use the find command with the exec expression.

9.6: 9-bash.sh

```
1 #!/bin/bash
2
3 # You don't need -R in chmod because we already have all the files we wish to change
   fetched.
4 find /home/jonathan/Pictures/ -type f -exec chmod 600 {} +
```

9.5 Using the find command responsibly

Before using the -exec expression for anything, you should just run the find command as is and check if the output of it are acutally the files and directories that you wish to remove. Otherwise you could alter something that you shouldn't have and end up bricking your system

Chapter 10

The grep command

10.1 What is grep?

Grep stands for Global Regular Expression Print. We don't have to get into regex for the tutorial whatsoever. You'll usually see it used with pipes '|' but we haven't seen those yet so we'll stick with grep. What grep does is that it looks for a specific string from a file or any output.

10.1: 10-bash.sh

```
1 #!/bin/bash
2
3 # Grep has a very basic structure
4 # command | string/pattern | path
5
6 grep Port /etc/ssh/ssh_config
```

10.2 Grep options

First, we have the '-v' option which usually means "verbose" or has to do with the command version. In this case, it's like a not operator. It will output all instances where the specified word does not appear.

10.2: Terminal

```
1 grep -v Port /etc/ssh/ssh_config
```

The next one is very useful as it tells us at which line the specified word was found.

10.3: Terminal

```
1 grep -n Port /etc/ssh/ssh_config
```

Then we have the '-c' option for count, this will tell us how many instances of the specified word were found

10.4: Terminal

```
1 grep -c Port /etc/ssh/ssh_config
```

If we wrote "port" instead of "Port", it wouldn't have worked since grep is case sensitive. But we can use the '-i' option or ignore case. very self explanatory

10.5: Terminal

```
1 grep -i port /etc/ssh/ssh_config
```

10.3 Search accross multiple files

We can search accross more than a single file for any keyword.

10.6: 10-bash.sh

```
1 #!/bin/bash
2
3 # Using the star wild card means search accross all files of current directory
4 grep find *
```

The command will tell you the file name at which the instance of the keyword was found.

There is also recursive search in case you want to search accross subdirectories:

10.7: Terminal

```
1 grep -r echo ./
```

10.4 Combining options

This is something general for all commands (I suppose). Instead of putting a billion flags in your command, you can use a single flag and put all your options there.

10.8: Terminal

```
1 grep -nri error /var/log
```